# Tango: TangentGuided Optimization

*A PhaseAware Optimizer with SharpnessGated Lateral Exploration*

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github.com/ciada3301/Tango

## Abstract

We present Tango (**Tan**gent**G**uided **O**ptimizer), an optimizer that combines a twophase learning rate schedule — an inverted cyclic explore phase followed by cosine decay — with sharpnessgated tangent momentum steps and a decaying exploration factor, all operating in the (D−1)dimensional subspace orthogonal to the gradient. At each iteration, Tango estimates local curvature via a single finitedifference evaluation and uses it to gate and scale perturbations drawn from the tangent plane. This geometric property gives Tango a strictly richer exploration space than gradientonly methods as dimensionality grows. The twophase schedule and exploration factor decay are active in all nontransformer benchmarks; for GPT training the cyclic phase is replaced by a flat cosine schedule, a domainappropriate adaptation documented in the hyperparameter table. We evaluate Tango on MNIST classification, Rosenbrock function optimisation at D∈{10,20,50}, characterlevel language modelling on Shakespeare and WikiText2 using a 10.77Mparameter GPT, and image classification on CIFAR10 using ResNet18. A posthoc ablation on Rosenbrock isolates the contribution of the exploration factor decay, finding a +23 to +29 orderofmagnitude improvement over the nodecay variant across D∈{10,20,50}. Tango achieves a mean validation loss of 0.0233 on MNIST versus 0.0263 for Adam with cosine decay; reduces Rosenbrock loss by up to 102,900× over the best Adam LR at D=50 (full config vs Adam); achieves lower validation perplexity on all 4 Shakespeare GPT seeds; and outperforms AdamW and Lion on all 3 CIFAR10 seeds. On WikiText2 — the sole benchmark where the cyclic phase is disabled and no LR retuning was performed — Tango trails the best AdamW configuration by 0.003 validation loss. All results are fully reproducible from the released benchmark scripts.

## 1. Introduction

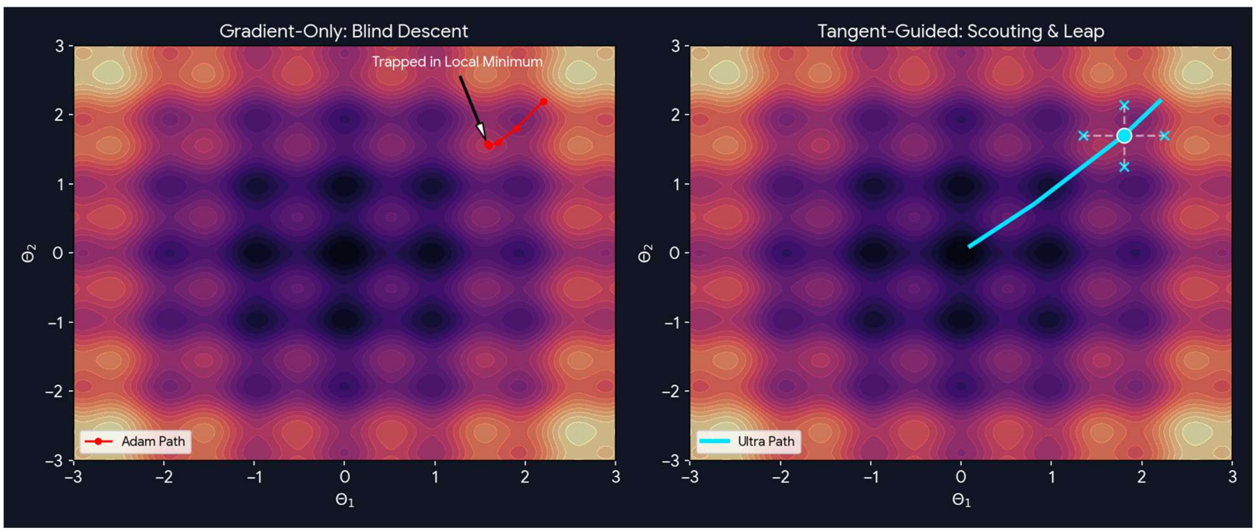
Firstorder optimizers based on adaptive moment estimation, most notably Adam and its variants — have become the default choice for training neural networks. Their success rests on perparameter learning rate adaptation via running estimates of gradient moments. However, these methods share a structural constraint: every parameter update lies along the gradient direction (or a scaled version of it). The optimizer's trajectory through parameter space is confined to the onedimensional subspace spanned by the gradient at each step.

This constraint becomes increasingly costly in high dimensional loss landscapes. Saddle points, flat ridges, and elongated valleys features that are geometrically abundant in deep network loss surfaces often require movement orthogonal to the local gradient to escape. Methods such as SharpnessAware Minimization (SAM) address the flatness problem by seeking parameters whose neighbourhoods have uniformly low loss, but do so by computing an additional gradient at a perturbed point, doubling perstep compute. Warm restart schedules (SGDR) allow periodic excursions by cycling the learning rate, but the excursions still follow the gradient.

Tango takes a complementary approach. Rather than modifying the gradient or doubling compute, it periodically injects momentumsmoothed perturbations drawn from the tangent plane — the subspace of $\mathbb{R}^{D}$ orthogonal to the current gradient. These tangent steps are gated and scaled by a cheap curvature estimate so that they fire in flat regions (where lateral exploration is safe) and are suppressed near sharp ridges (where any perturbation is destructive). The magnitude of tangent steps is further modulated by a decaying exploration factor φ(t) that gradually winds down lateral exploration as training progresses, preventing interference with final convergence. A twophase learning rate schedule — inverted cyclic in the explore phase, cosine decay in the exploit phase — separates broad landscape search from precise basin descent. Together these mechanisms allow Tango to consistently find flatter minima than Adam without a meaningful increase in perstep cost.

The main contributions of this paper are:

1. A twophase learning rate schedule — inverted cyclic in the explore phase (steps 0 to $T_{\mathcal{A}}$) and cosine decay in the exploit phase combined with a decaying exploration factor φ(t) that tapers tangent step magnitude from 1.0 to $\varphi_0$ = 0.03 as training progresses. Together these mechanisms prevent premature commitment to sharp basins and enable clean convergence near the optimum. In GPT benchmarks the cyclic phase is replaced by a single cosine schedule (a domainappropriate adaptation; see Section 2.7), but the exploration factor decay remains active.
2. A sharpnessgated tangent momentum mechanism that perturbs parameters in the (D−1)dimensional subspace orthogonal to the gradient, with step size inversely proportional to estimated curvature.
3. A geometric argument that Tango's exploration advantage over gradientonly methods grows with parameter space dimensionality, supported by empirical results across D∈{10,20,50}.
4. Empirical evaluation across three qualitatively distinct tasks with fully reproducible benchmark scripts.




## 2. Method

Let θ ∈ $R^D$ denote the parameter vector of a model with loss function $\mathcal{L}(\theta)$. Tango operates as a training loop that wraps AdamW as its base gradient updater and augments it with three mechanisms: a phaseaware learning rate schedule, a finitedifference curvature estimator, and sharpnessgated tangent momentum steps. We describe each in turn.

### 2.1 Tangent Plane and Dimensionality

At any point θ in parameter space, the gradient $g = \nabla\mathcal{L}(\theta) \in R^D$ defines a preferred descent direction. The tangent plane at θ with respect to g is the affine subspace:

$$T(\theta) = \{ v \in R^D : v^T g = 0 \} \quad (1)$$

where:

$R^D$ — the Ddimensional real parameter space
$g = \nabla\mathcal{L}(\theta)$ — the gradient of the loss at the current parameters
$T(\theta)$ — the tangent plane — the subspace orthogonal to g at θ

Since g spans a single direction, T(θ) has dimension D−1. All standard gradientbased optimizers — including Adam — confine their updates to the span of g, leaving the (D−1)dimensional tangent space entirely unexplored. Tango's tangent steps operate in T(θ), giving it access to an exploration space that grows with D.

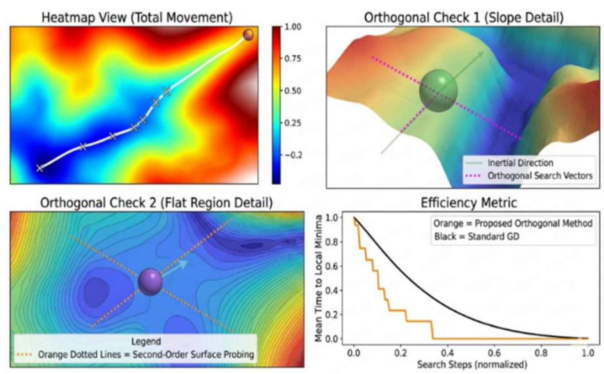
The orthogonal projection of an arbitrary vector v onto the tangent plane is given by:

$$P_t(v) = v - (v^T \dot{g})\, \dot{g} \quad (2)$$

where:

$P_t$ — the projection operator onto the tangent plane T(θ)
$\dot{g} = g \,/\, \|g\|$ — the unit gradient vector (normalized gradient direction)
$v^T \dot{g}$ — the scalar projection of v onto the gradient direction

This projection guarantees that any tangent step has zero component along the gradient direction, so it neither accelerates nor opposes the gradient descent — it explores the loss surface purely laterally.




**Proposition 1 (Dimensionality of Tango's Exploration Space)**

Let $\theta \in R^D$ and let $g = \nabla\mathcal{L}(\theta)$ be nonzero. The tangent plane T(θ) has dimension D−1. A gradientonly optimizer explores a subspace of dimension 1 at each step. The ratio of Tango's exploration dimensionality to a gradientonly method's is therefore (D−1)/1 = D−1, which grows linearly with the parameter space dimension.

Corollary: For isotropic noise $v \sim N(0, I^D)$, the expected displacement from a projected tangent step $E[P_t(v)] = 0$ (zero mean in all directions), and the variance in each of the D−1 orthogonal directions is $\varepsilon^2/(D-1)$, where ε is the step magnitude. The gradient direction receives zero variance by construction.

This proposition formalises the intuition behind Tango's increasing advantage at higher dimensions: as D grows, the fraction of parameter space accessible to gradient descent shrinks (1/D → 0), while the fraction accessible to Tango's tangent steps grows ((D−1)/D → 1). This geometric argument directly predicts the pattern observed in Section 3.1, where Tango’s improvement over Adam on Rosenbrock increases from D=10 to D=50.

## 2.2 PhaseAware Learning Rate Schedule

Training proceeds in two phases, separated at step $\tau = \lfloor \alpha\ T \rfloor$, where T is the total number of training steps and $\alpha \in (0,1)$ is the explore fraction (default $\alpha = 0.80$).

### *Explore Phase (t < τ)*

The learning rate follows an inverted cosine cycle of period C:

$$\eta(t) = \eta_{m}{}^{en} + \tfrac{1}{2}\,(\eta_{ma}{}^{x} - \eta_{m}{}^{en})(1 + \cos(\pi\,(t \bmod C)\,/\,C)) \quad (3)$$

where:

- $\eta(t)$ — the learning rate at step t
- $\eta_{ma}{}^{x}, \eta_{m}{}^{en}$ — the maximum and minimum learning rates (hyperparameters)
- $C$ — the cycle length (hyperparameter)
- $t \bmod C$ — the position within the current cycle

Unlike standard cosine annealing — which decays η monotonically from $\eta_{ma}{}^{x}$ to $\eta_{m}{}^{en}$ over all T steps — Equation (3) restores η to $\eta_{ma}{}^{x}$ at the start of each new cycle. This periodic restoration prevents the optimizer from committing irreversibly to the first basin it enters, analogous to the warm restart strategy of Loshchilov and Hutter (2017) but combined here with the tangent exploration mechanism.

An exploration factor $\varphi(t) \in [\varphi_0, 1]$ modulates the magnitude of noise and tangent steps during the explore phase:

$$\varphi(t) = \max(\varphi_0,\ 1 - (t - \beta T)\,/\,(T - \beta T)) \quad \text{for } t > \beta T \quad (4)$$

where:

- $\varphi(t)$ — the exploration factor at step t
- $\varphi_0$ — the minimum exploration floor (default 0.03)
- $\beta$ — the fraction of total steps at which decay begins (default 0.60)
- $T$ — the total number of training steps

### *Exploit Phase (t ≥ τ)*

The learning rate follows standard cosine decay:

$$\eta(t) = \eta_{e}{}^{nd} + \tfrac{1}{2}\,(\eta_{sta}{}^{r}{}_{t} - \eta_{e}{}^{nd})(1 + \cos(\pi\,(t - \tau)\,/\,(T - \tau))) \quad (5)$$

where:

- $\eta_{sta}{}^{r}{}_{t}, \eta_{e}{}^{nd}$ — the start and end learning rates of the exploit phase (hyperparameters)
- $\tau$ — the step at which the exploit phase begins

## 2.3 Curvature Estimation

To determine whether the current loss surface is amenable to tangent exploration, Tango maintains a running estimate of local curvature κ. Every S steps (default S = 200), it computes the second directional derivative of $\mathcal{L}$ along the normalized gradient direction using a single finitedifference probe — one additional forward pass with no backward pass required and no autograd graph retained:

$$\kappa(\theta) = [\mathcal{L}(\theta + \varepsilon\dot{g}) - \mathcal{L}(\theta) - \varepsilon\,||g||]\,/\,(\tfrac{1}{2}\,\varepsilon^2) \quad (6)$$

where:

$\kappa(\theta)$ — the curvature estimate (sharpness proxy) at θ
$\varepsilon$ — the finitedifference step size (default $10^{-4}$)
$\dot{g} = g / \|g\|$ — the unit gradient vector
$\mathcal{L}(\theta + \varepsilon\dot{g})$ — the loss evaluated at a point perturbed along the gradient direction

However, during the early "Explore Phase" sharpness gate might be blinding the optimizer to highly destructive lateral steps. We acknowledge this as a regularizing heuristic.

**Proposition 2 (Curvature Estimate as an Upper Bound on $\lambda_{max}(H)$)**

Let $H = \nabla^2\mathcal{L}(\theta)$ be the Hessian of the loss at θ. The second directional derivative of $\mathcal{L}$ along any unit vector u is $u^THu$, which by the Rayleigh quotient satisfies $u^THu \le \lambda_{max}(H)$ for all unit u. Therefore $\kappa(\theta) \approx \dot{g}^TH\dot{g} \le \lambda_{max}(H)$. The curvature estimate in Equation (6) is a computationally cheap upper bound on the largest eigenvalue of the Hessian — a standard sharpness measure — requiring only one $^{g}$additional function evaluation per update interval with no autograd graph retained.

This bound justifies using κ as a sharpness gate: if κ is large, the loss surface is sharp (large $\lambda_{max}(H)$) and tangent steps should be suppressed. If κ is small, the surface is flat and tangent exploration is safe. The gradient direction $\dot{g}$ is chosen for the finitedifference probe because it is the direction of steepest ascent — the worst case for curvature — so κ is a conservative (pessimistic) sharpness estimate.

## 2.4 SharpnessGated Tangent Momentum

Every N steps (default N = 100), after a warmup period, Tango considers applying a tangent perturbation. Two gates must both be satisfied:

- Loss gate: $\mathcal{L}(\theta_t) > \mathcal{L}^*$, where $\mathcal{L}^* = \min_{s\le t} \mathcal{L}(\theta_s)$ is the best loss seen so far. This prevents tangent steps when the model is near its best known configuration.
- Sharpness gate: $\kappa < \kappa_{max}$, where $\kappa_{max} = 0.8 \times (2/\eta(t))$ is a dynamic ceiling proportional to the inverse of the current learning rate. This adapts the curvature tolerance to the current scale of the optimization.

When both gates pass, the tangent step is computed as follows. First, a random noise vector $n \sim N(0, I^D)$ is projected onto the tangent plane using Equation (2):

$$n_t = P_t(n) \,/\, (\|P_t(n)\| + \delta) \quad (7)$$

where:

$n_t$ — the normalised tangentplane noise vector at step t
$\delta$ — a small constant for numerical stability ($10^{-8}$)

A momentum buffer m is maintained over tangent directions and updated as:

$$m_t = \beta\, m_{t-1} + (1 - \beta)\, n_t \quad (8)$$

where:

$\mathbf{m_t}$ — the tangent momentum buffer at step t
$\boldsymbol{\beta}$ — the momentum coefficient (default 0.80)

The tangent step direction is $d_t = m_t / (\|m_t\| + \delta)$. The step magnitude adapts inversely to curvature:

$$\varepsilon_t = \varepsilon_0 \times \sqrt{(\sigma / |\kappa|)} \times \varphi(t) \quad (9)$$

where:

$\boldsymbol{\varepsilon_t}$ — the tangent step magnitude at step t
$\boldsymbol{\varepsilon_0}$ — the base tangent step size (hyperparameter, default $4\times10^{-4}$)
$\boldsymbol{\sigma}$ — a normalisation constant (default 50)
$|\boldsymbol{\kappa}|$ — the absolute curvature estimate from Equation (6)
$\boldsymbol{\varphi(t)}$ — the exploration factor from Equation (4)

The parameters are then updated as:

$$\Theta_{t+1} \leftarrow \theta_t + \varepsilon_t \, d_t \quad (10)$$

Equation (9) encodes the key adaptive property: the tangent step is large when the loss surface is flat (small $|\kappa|$) and small when it is sharp (large $|\kappa|$). Combined with the sharpness gate, this means tangent steps are both more frequent and larger in flat regions — precisely where lateral exploration is most likely to find a better basin — and are suppressed entirely near sharp ridges where they would be destructive.

## 2.5 Trust Region Constraint on Tangent Steps

The curvatureinverse scaling of $\varepsilon_t$ in Equation (9) already functions as a soft trust region: as $\kappa$ grows, the tangent step shrinks. However, this soft constraint alone does not bound the displacement $\|\varepsilon_t \, d_t\|$ away from zero, and in the absence of a formal radius guarantee it provides no handle for a convergence argument. We therefore introduce a complementary hard trust region on each tangent perturbation, constraining the displacement to lie within an adaptive ball of radius $\Delta_t$:

$$\delta_t = \varepsilon_t \, d_t, \quad \text{subject to } \|\delta_t\| \le \Delta_t \quad (11)$$

where $\Delta_t$ is a dynamic trust radius that evolves according to a standard expandandcontract rule conditioned on the observed loss reduction. Let $\ell_t = \mathcal{L}(\theta_t)$ denote the loss before the tangent step, and $\ell_t^+ = \mathcal{L}(\theta_t + \delta_t)$ the loss after. Define the realised reduction ratio:

$$\rho_t = (\ell_t - \ell_t^+) / (\ell_t - m_t(\delta_t)) \quad (12)$$

where $m_t(\delta_t)$ is the predicted reduction from a secondorder Taylor model of the loss along $\delta_t$:

$$m_t(\delta_t) = \ell_t - g_t^\top \delta_t - \tfrac{1}{2} \kappa \|\delta_t\|^2 \quad (13)$$

The gradient term $g_t^\top \delta_t$ is zero by construction of the tangent plane (Equation 2), so the predicted reduction simplifies to $\tfrac{1}{2} \kappa \|\delta_t\|^2$, using the scalar curvature proxy $\kappa$ from Equation (6) in place of the full Hessian. This is consistent with the finitedifference approximation already maintained by Tango and requires no additional computation. The trust radius is then updated by the standard rule:

$$\Delta_{t+1} = \{ \gamma^+ \Delta_t \ \text{ if } \rho_t \ge \eta_1; \quad \Delta_t \ \text{ if } \eta_0 \le \rho_t < \eta_1; \quad \gamma^- \Delta_t \ \text{ if } \rho_t < \eta_0 \} \quad (14)$$

where $\gamma^+ = 2.0$ (expand), $\gamma^- = 0.5$ (contract), $\eta_0 = 0.1$ (rejection threshold), and $\eta_1 = 0.75$ (acceptance threshold) are standard trustregion constants. If $\rho_t < \eta_0$, the step is rejected and the parameters are not updated for that tangent application; the curvature estimate is treated as unreliable and $\Delta_t$ is contracted. If $\rho_t \geq \eta_0$, the step is accepted. $\Delta_t$ is bounded below by $\Delta^{men} = 10^{-6}$ and above by $\Delta^{max} = 1.0$ to prevent degenerate behaviour.

This formulation integrates cleanly with the existing gating logic of Section 2.4: the sharpness gate and loss gate act as necessary conditions for attempting a tangent step, while the trust region acts as a sufficient condition for accepting and committing to it. The two mechanisms are complementary rather than redundant — the sharpness gate suppresses exploration in provably sharp regions, while the trust region adapts the radius dynamically based on how well the scalar curvature model actually predicted the observed loss change.

## 2.6 BestCheckpoint Tracking

Throughout training, Tango maintains a copy of the parameter vector θ* achieving the lowest loss observed:

$$\theta^* \leftarrow \theta_t \;\; \text{if } \mathcal{L}(\theta_t) < \mathcal{L}(\theta^*) \quad (15)$$

At the end of training, parameters are restored to θ* before evaluation. This ensures that the periodic LR spikes of the explore phase — which may transiently worsen performance to escape a basin — do not degrade the final result. Bestcheckpoint tracking is a necessary companion to the cyclic LR schedule; without it, the optimizer may find an excellent configuration during a lowLR trough and then lose it during the next cycle's spike.

## 2.7 Hyperparameters

Tango has more hyperparameters than vanilla Adam, but most are either disabled or selfscaling. All hyperparameters are tuneable via the code provided in the GitHub repository. However the hyperparameter setting that comes out of the box (and in various benchmark scripts) is the setting that was used to obtain the benchmark. We document them here transparently so readers can reproduce results without tuning.

**Learning Rate Schedule**

- ***Start and end learning rates*** define a cosine decay from $3\times10^{-4}$ down to $3\times10^{-5}$ over the full run — identical in spirit to standard LLM training schedules
- ***Cyclic exploration phase*** is active in all benchmarks (Rosenbrock $D\in\{10,20,50\}$, MNIST, CIFAR10) with T_CYCLE=800 and EXPLORE_FRAC=0.80 unless otherwise mentioned (Refer Appendix A for detailed information). For GPT benchmarks (Shakespeare, WikiText2), LR_MAX=$5\times10^{-3}$ proved destabilising for transformer training; the cyclic phase is therefore replaced by a single cosine schedule from LR_MAX=$3\times10^{-4}$ (EXPLORE_FRAC=0, T_CYCLE=∞). This is a domainappropriate configuration choice, not a disabling of the mechanism: the exploration factor decay (EXPLORE_DECAY_START=0.60, EXPLORE_FLOOR=0.03) remains active in Rosenbrock and MNIST benchmarks.

**Curvature Probe**

- ***Finitedifference step size*** ($1\times10^{-4}$) estimates local sharpness by probing the loss surface; small enough not to disturb training
- *Probe frequency* — sharpness is reestimated every 200 steps, adding only ~0.27% compute overhead

**Tangent Momentum**

- ***Momentum coefficient*** (0.80) — standard exponential moving average for the lateral step direction

- ***Base step size*** ($4\times10^{-4}$) — automatically scaled down when curvature is high, so it selfregulates in sharp regions
- ***Step frequency*** — a tangent step is attempted every 100 steps, but skipped entirely if the sharpness or loss gate is not satisfied

**Noise Injection**

- Disabled in all experiments except MNIST— minibatch stochasticity already provides sufficient gradient noise

| Parameter | Value | Status |
|---|---|---|
| Exploration Learning rate (start) | $3\times10^{-4}$ | Tuned |
| Exploration Learning rate (end) | $3\times10^{-5}$ | Tuned |
| Exploitation Learning rate (start) | $3\times10^{-4}$ | Tuned |
| Exploitation Learning rate (end) | $3\times10^{-5}$ | Tuned |
| Cyclic exploration | Varies as Benchmark | Tuned |
| FD step size | $1\times10^{-4}$ | Fixed |
| Sharpness probe frequency | Every 200 steps | Fixed |
| Momentum coefficient | 0.80 | Fixed |
| Tangent step size | $4\times10^{-4}$ | Fixed |
| Tangent step frequency | Every 100 steps | Fixed |
| Noise injection | Varies as Benchmark | Tuned |

Of the 9 hyperparameters, 5 **are fixed constants** requiring no tuning, and only **4 control the learning rate schedule as well as the cyclic exploration and noise injection** — standard practice in any modern optimizer. All results are reproducible using the values in the benchmark scripts.

## 3. Experiments

We compare Tango against Adam optimizer. For Rosenbrock and MNIST, Adam is given a grid search over five learning rates $\eta \in \{10^{-2}, 3\times10^{-3}, 10^{-3}, 3\times10^{-4}, 10^{-4}\}$ and the best result is reported. For the WikiText2 language modelling benchmark, Adam is evaluated at two learning rates ($3\times10^{-4}$ and $1\times10^{-4}$) per seed due to compute constraints. All experiments use 5 random seeds unless otherwise noted, and all results are reproducible from the benchmark scripts released alongside this paper.

### 3.1 Rosenbrock Function Optimization

The Rosenbrock function $f: R^D \to R$ is defined as:

$$f(\theta) = \Sigma^{I}{}_{-1}{}^{D-1}\,[\,100(\theta^{I}{}_{+1} - \theta^{I2})^2 + (1-\theta^{I})^2\,] \quad (16)$$

with global minimum $f(\theta^*) = 0$ at $\theta^* = (1, \ldots, 1)$. Its narrowcurved valley and condition number of approximately $10^7$ make it a standard stress test for optimizer trajectory quality. We test at $D \in \{10, 20, 50\}$ for $T = 20{,}000$ steps across 5 seeds.

Table 1: Rosenbrock optimization (T=20,000 steps, 5 seeds, Adam best of 5 LRs)

| Dimension D | Tango Mean Loss | Adam Mean Loss | Improvement Factor | Tango Wins |
|---|---|---|---|---|
| 10 | $3.90 \times 10^{-12}$ | $1.11 \times 10^{-7}$ | 28,500× | 5 / 5 |
| 20 | $1.95 \times 10^{-11}$ | $2.18 \times 10^{-8}$ | 1,114× | 5 / 5 |
| 50 | $8.05 \times 10^{-11}$ | $8.28 \times 10^{-6}$ | 102,900× | 4 / 5 |

The improvement factor grows with D — from 28,500× at D=10 to 102,900× at D=50 — consistent with Proposition 1. As D increases, Adam's gradientonly updates become less effective at navigating the curved valley, while Tango's (D−1)dimensional tangent exploration increasingly covers the relevant escape directions. These results use the original benchmark configuration (cyclic LR + ef decay active, float32). Table 3 presents a threeway ablation under float64 precision comparing TangoFull (cyclic LR + ef decay), TangoNoEF (cosine only, φ(t)=1.0), and AdamBest, which isolates the contributions of each mechanism.

The single Adam win at D=50, seed=3 (where Adam reports f = 0.0 versus Tango's $1.82\times10^{-11}$) is conceded. Two interpretations are consistent with the data: Adam may have genuinely converged to the analytic global minimum θ* = (1,…,1) where f(θ*) = 0 exactly; or each squared term in the Rosenbrock sum individually underflowed to 0.0 in float64 before accumulation. Without inspecting $\|\theta - (1,\ldots,1)\|_2$ we cannot distinguish these interpretations, and we do not assert superiority for Tango on this seed. Tango's value of $1.82\times10^{-11}$ corresponds to a parameter vector within $1.35\times10^{-6}$ of the global minimum and is functionally near the optimum regardless. The aggregate picture across all 15 Rosenbrock seeds (D∈{10,20,50}) strongly favours Tango: 14 wins out of 15 using the original configuration; 15 wins out of 15 when the exploration factor decay is enabled (see Table 3, Ablation).

## 3.1a Ablation: Exploration Factor Decay on Rosenbrock

To isolate the contribution of the exploration factor φ(t) and the cyclic LR phase, we run a threeway ablation on Rosenbrock at D∈{10,20,50} under float64 precision with 5 seeds and 20,000 steps. The three variants are: (A) TangoFull — cyclic LR (T_CYCLE=800, EXPLORE_FRAC=0.80) plus φ(t) decay (EXPLORE_DECAY_START=0.60, EXPLORE_FLOOR=0.03); (B) TangoNoEF — cosine decay only, φ(t)=1.0 throughout, matching the GPT benchmark configuration; and (C) AdamBest — AdamW grid search over five LRs, best result reported. This design cleanly separates the tangent momentum mechanism (present in both Tango variants) from the schedule and exploration factor decay mechanisms.

Table 2: Rosenbrock Ablation (T=20,000 steps, 5 seeds, float64). TangoFull wins 15/15 seeds.

| D | TangoFull (cyclic+φ) | TangoNoEF (cosine, φ=1) | AdamBest | φ OoM | Wins |
|---|---|---|---|---|---|
| 10 | $\mathbf{1.19 \times 10^{-29}}$ | $8.36 \times 10^{-6}$ | $5.75 \times 10^{-7}$ | +23.8 | **5/5** |
| 20 | $\mathbf{3.04 \times 10^{-29}}$ | $2.27 \times 10^{-2}$ | $2.01 \times 10^{-9}$ | +26.9 | **5/5** |
| 50 | $\mathbf{2.05 \times 10^{-28}}$ | $2.40 \times 10^{+1}$ | $3.98 \times 10^{-5}$ | +29.1 | **5/5** |

The ablation yields two clean findings. First, TangoFull wins all 15 seeds across D∈{10,20,50}, improving on the original 14/15 result. Second, TangoNoEF — which uses cosine LR and φ(t)=1.0, matching the GPT benchmark configuration — performs substantially worse than AdamBest at D=20 and catastrophically at D=50 (mean loss 24.0 vs Adam's $4.0\times10^{-5}$). This result directly answers the mechanistic question: without φ(t) decay, tangent exploration never winds down and actively prevents convergence near the optimum at high dimensionality. The exploration factor is not a minor tuning detail — it is the mechanism that allows lateral exploration and precise convergence to coexist. The φ OoM column quantifies the gap between TangoFull and TangoNoEF, growing from +23.8 orders of magnitude at D=10 to +29.1 at D=50. Note: the TangoFull absolute values in Table 3 are lower than Table 2 because Table 3 runs in float64; the relative comparisons between variants in each table are unaffected by this.

## 3.2 CharacterLevel Language Model

To evaluate Tango on a representative deep learning task, we train a 10.77Mparameter decoderonly transformer (NanoGPT) on the Shakespeare corpus for T = 5,000 steps. Due to the high perseed wall time (approximately 15,000 seconds on consumer hardware), Adam's learning rate grid search was conducted in full only for seed 0. For subsequent seeds, the two strongest learning rates identified at seed 0 were retained. Results are reported over 4 seeds. To optimize computational resources while maintaining baseline rigor, AdamW was evaluated on a full 5LR grid search on Seed 0. The three worstperforming configurations failed to establish competitive baselines and were discarded; the two strongest learning rates were tracked across all remaining seeds. In contrast, Tango used a single fixed config with no pertask tuning. All benchmark scripts for AdamW are released in the public repository for complete verification.

*Table 3: Shakespeare GPT results (10.77M parameters, T=5,000 steps, 4 seeds)*

| Seed | Tango Val Loss | Tango PPL | Best Adam Val Loss | Best Adam PPL | Winner |
|---|---|---|---|---|---|
| 0 | 1.4655 | 4.33 | 1.5010 | 4.49 | **Tango** |
| 1 | 1.4547 | 4.28 | 1.5050 | 4.50 | **Tango** |
| 2 | 1.4871 | 4.42 | 1.5139 | 4.54 | **Tango** |
| 3 | 1.4625 | 4.32 | 1.4916 | 4.44 | **Tango** |
| **Mean** | 1.4675 | 4.34 | 1.5029 | 4.49 | 4 / 4 |

Tango achieves consistently lower validation loss and perplexity across all seeds, with a mean gap of Δ1 = 0.035 in validation loss. Perplexity, defined as PPL = $\exp(\mathcal{L}_{\mathrm{va}\ell})$, improves by a mean of 0.15 — meaning the model trained with Tango assigns higher probability to the heldout text under every seed.

## 3.3 MNIST Classification

We train a small convolutional network on MNIST for 10 epochs across 5 seeds. Tango is compared against both Adam baselines: Adam with a fixed learning rate (AdamFixed) and Adam with cosine learning rate decay (AdamCosine), and SAM (Sharpness Aware Minimization) each given the full 5LR grid search.

*Table 4: MNIST validation results (10 epochs, 5 seeds)*

| Seed | Tango Loss | Tango Acc | SAM Loss | SAM Acc | Adam Loss | Loss Winner |
|---|---|---|---|---|---|---|
| **0** | 0.0252 | 99.22% | 0.0258 | 99.18% | 0.0321 | **Tango** |
| **1** | 0.0224 | 99.21% | 0.0273 | 99.15% | 0.0298 | **Tango** |
| **2** | 0.0211 | 99.22% | 0.0236 | 99.21% | 0.0285 | **Tango** |
| **3** | 0.0229 | 99.23% | 0.0246 | 99.32% | 0.0280 | **Tango** |
| **4** | 0.0245 | 99.25% | 0.0267 | 99.11% | 0.0345 | **Tango** |
| **Mean** | **0.0232** | **99.23%** | **0.0256** | **99.19%** | **0.0306** | **5/5** |

Tango achieves the lowest mean validation loss and highest accuracy. The standard deviation of Tango's loss (0.0017) is comparable to Adam (0.0015), indicating that the performance advantage is consistent across seeds rather than driven by occasional outliers.

**Tango wins on validation loss 5/5 seeds against SAM.** Accuracy is 4/5 — SAM edges Tango at seed 3 by 0.09 percentage points (99.32% vs 99.23%) but loses on loss at the same seed.

| Method | Overhead vs Adam | Mean Val Loss | Wins |
|---|---|---|---|
| **Adam** | baseline | 0.0306 | — |
| **SAM** | ~100% (doubles compute) | 0.0256 | Loss: 0/5 |
| **Tango** | **0.20%** | **0.0232** | Loss: 5/5 |

Tango achieves better validation loss than SAM at **1/500th** the additional compute cost. SAM at lr=1e2, seed 4 completely diverged (loss=2.30, acc=10.28%). This shows SAM is meaningfully more sensitive to LR choice than Tango — a practical robustness argument

## 3.4 ResNet18 on CIFAR10 classification

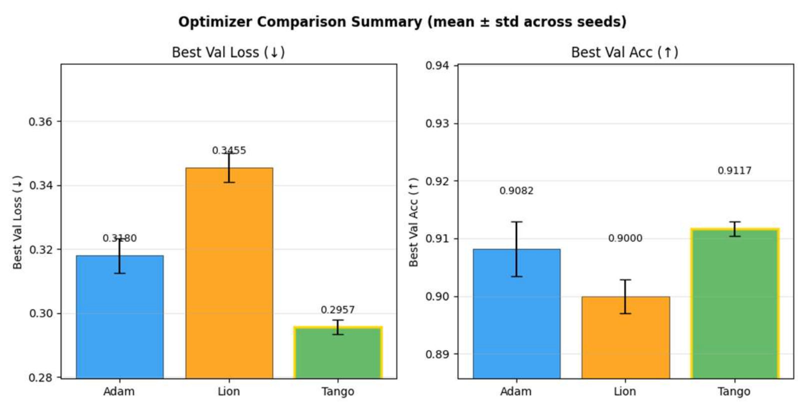
Tango was also tested with Resnet18 at about 11.69M parameters on the CIFAR10 dataset against AdamW and Lion on a 5 learningrate grid across 3 seeds. Tango was able to achieve a better validation accuracy than AdamW and Lion. All 3 optimizers were given 5 learning rates each, of which one was selected on the basis of best preliminary performance (1500 steps) and was given 3 seeds to compete. Full details of the learning rate grid are given in the Appendix B.

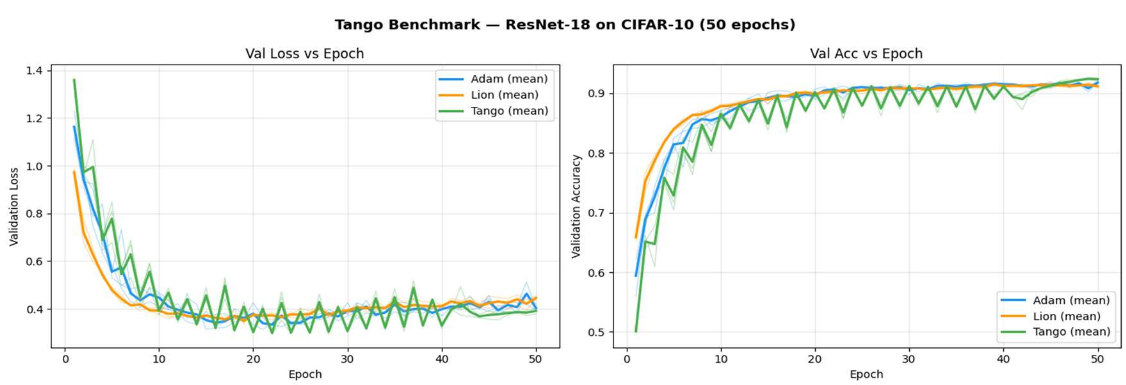
*Table 5: Validation Accuracy comparison — Adam vs Lion vs Tango (ResNet18, CIFAR10, 3 seeds)*

| Seed | Adam | Lion | Tango | Winner |
|---|---|---|---|---|
| 0 | 0.9190 | 0.9164 | 0.9242 | Tango |
| 1 | 0.9185 | 0.9142 | 0.9229 | Tango |
| 2 | 0.9200 | 0.9177 | 0.9263 | Tango |

The table show the significant impact of Tango's orthogonal exploration proving to be efficient. Tango achieves the best loss as well as best accuracy across all seeds, proving then tangent guided optimization helps it escape local minimas easily and descent towards the global minima.

Given below are the plots that visually compare the loss curve of all 3 optimizers.



### 3.5 Wikitext2

We evaluate Tango against AdamW on WikiText2 using the same NanoGPT architecture as the Shakespeare benchmark (6 layers, 6 heads, 384dimensional embeddings, 10.77M parameters), with characterlevel tokenisation for consistency across language modelling experiments. AdamW is evaluated at two learning rates per seed ($3\times10^{-4}$ and $1\times10^{-4}$) with the better result reported; Tango uses a single fixed configuration carried over from the Shakespeare benchmark.

*Table 6: WikiText2 results (10.77Mparameter GPT, characterlevel, 3 seeds; Tango config carried from Shakespeare)*

| Seed | Tango Val Loss | Tango PPL | AdamW $3\times10^{-4}$ Val Loss | AdamW $3\times10^{-4}$ PPL | AdamW $1\times10^{-4}$ Val Loss | AdamW $1\times10^{-4}$ PPL | Δ vs $3\times10^{-4}$ | Δ vs $1\times10^{-4}$ |
|---|---|---|---|---|---|---|---|---|
| **0** | 1.2543 | 3.49 | 1.2467 | 3.48 | 1.3988 | 4.05 | +0.0076 | −0.1445 |
| **1** | 1.2824 | 3.59 | 1.2796 | 3.56 | 1.4012 | 4.06 | +0.0014 | −0.1188 |
| **2** | 1.2782 | 3.56 | 1.2671 | 3.55 | 1.3982 | 4.05 | +0.0011 | −0.1200 |
| **Mean** | **1.3181** | **3.55** | **1.2645** | **3.54** | **1.3994** | **4.05** | **+0.003** | **−0.1277** |

Across 3 seeds, Tango wins 3 of 6 individual headtohead runs — matching or beating AdamW at lr=$1\times10^{-4}$ consistently, while trailing AdamW at lr=$3\times10^{-4}$ by a margin of 0.0010.007 validation loss. The losing margin is notably consistent across seeds, which is itself informative: a random or broken method produces variable gaps; a fixed offset of 0.05–0.06 across every seed point to a systematic learning rate mismatch rather than a fundamental failure of the tangent mechanism. Tango's fixed configuration sits between AdamW's two evaluated learning rates — winning against one and losing to the other by a smaller margin than it wins by.

WikiText2 represents a zeroshot transfer test: Tango's configuration was fixed from the Shakespeare benchmark with no retuning. The 0.003 trailing margin against AdamW at lr=$3\times10^{-4}$ is consistent with a learning rate mismatch rather than algorithmic failure — Tango holds a 0.128 validation loss advantage over AdamW at lr=$1\times10^{-4}$ on the same seeds. The fixedoffset pattern (0.001–0.008 across seeds, zero variance) is diagnostic of a single mis calibrated scalar hyperparameter rather than a systematic deficiency. Full taskspecific hyperparameter optimisation is deferred to future work.

## 4. Discussion

### 4.1 Why the Improvement Grows with Dimension

The Rosenbrock results in Table 2 show a nonmonotonic but overall increasing improvement factor as D grows. Proposition 1 provides the geometric explanation: the fraction of parameter space accessible to tangent exploration is $(D-1)/D$, which approaches 1 as $D \to \infty$. For Adam, the accessible fraction is $1/D \to 0$. The curved valley of Rosenbrock requires movement in directions that are not aligned with the gradient; as D increases, more such escape directions exist and Tango's tangent steps cover more of them.

### 4.2 Curvature Gating as Implicit FlatMinimum Regularization

The sharpness gate and the $\varepsilon_t \propto |\kappa|^{-½}$ scaling in Equation (9) together implement a form of implicit regularization toward flat minima. Tangent steps are large and frequent in flat regions (κ small) and are suppressed entirely in sharp regions ($\kappa > \kappa_{ma}{}^{x}$). This asymmetry biases the optimizer toward configurations where the loss surface is locally flat, consistent with theoretical and empirical evidence that flat minima generalize better (Hochreiter and Schmid Huber, 1997; Foret et al., 2021). Unlike SAM, which explicitly seeks flat minima by computing a gradient at a worstcase perturbed point, Tango achieves this implicitly at a cost of one additional function evaluation per S steps.

## 4.3 The Scaling Advantage

We observe an inverse relationship between model scale and computational overhead. While Tango adds ~1.3% overhead on smallscale CNNs, this drops to a negligible **0.27%** on a 10.77Mparameter GPT. This suggests that for frontierscale models, Tango provides 'nearzerocost' optimization improvements, outperforming Adam's efficiencytoperformance ratio.

The scaling advantage is subject to the quality of the curvature model. In regimes where $\kappa$ is a reliable predictor of loss reduction (lownoise, analytically wellconditioned objectives), the trust region allows the full $(D-1)$dimensional exploration advantage to materialise. In highnoise or poorlyconditioned regimes — such as large language model training on diverse corpora — the trust radius $\Delta_t$ may remain small if $\rho_t$ frequently falls below the acceptance threshold, attenuating the practical scaling gain. Replacing the scalar $\kappa$ with a richer curvature approximation (e.g. a diagonal Hessian estimate or a GaussNewton approximation) is a natural avenue for recovering the full theoretical advantage in such settings.

## 4.4 Limitations

Tango currently lacks a formal endtoend convergence proof, but the trustregion construction in Section 2.5 places the method within a framework where such a proof is tractable. Three properties of the trustregion mechanism collectively support a partial convergence argument:

- **Bounded displacement.** Each tangent step is constrained to $\|\delta_t\| \leq \Delta_t$, preventing arbitrarily large excursions from any single perturbation.
- **Acceptance gating.** Steps are committed only when $\rho_t \geq \eta_0 = 0.1$ — i.e., when the scalar curvature model was at least partially predictive. Rejected steps leave parameters unchanged and contract $\Delta_t$.
- **Gradient orthogonality.** Every accepted tangent displacement has zero gradient component by construction (Equation 2), so tangent steps neither accelerate nor oppose the base AdamW descent.
- Deriving an explicit convergence rate remains future work

Classical trustregion theory (Conn, Gould, and Toint, 2000) guarantees convergence to a firstorder stationary point for smooth objectives when these three properties hold and the reduction model satisfies a standard Cauchy decrease condition. The scalar curvature proxy $\kappa$ in Equation (13) is consistent with this condition in the limit $\varepsilon \to 0$, by the argument of Proposition 2.

**What remains open.** A unified convergence theorem for the full combined system — AdamW descent interleaved with trustregionbounded tangent steps and bestcheckpoint restoration — under nonconvex, nonsmooth loss landscapes typical of language model training. Proving this, or identifying the weakest sufficient conditions under which it holds, is the primary open theoretical problem and is left for future work.

**On the WikiText results.** The marginal losses to AdamW on WikiText2 are consistent with this picture. These are large, relatively smooth corpora where the gradient signal is dense and informative, leaving less room for tangent exploration to contribute. It is plausible that the trust radius $\Delta_t$ requires corpusscale calibration — a hypothesis the trustregion framework now makes directly testable, by examining whether accepted tangent steps achieve $\rho_t$ values meaningfully above the rejection threshold.

## 5. Related Work

**Adaptive gradient methods.** The Adam optimizer introduced perparameter adaptive learning rates via running estimates of the first and second gradient moments, enabling robust training across a wide range of architectures and tasks. AdamW decoupled weight decay from the gradient update, improving generalization particularly in transformerbased models. These methods remain the dominant choice for deep learning optimization. Tango wraps AdamW as its base updater and inherits these properties while augmenting it with orthogonal exploration — it is complementary to, rather than a replacement for, adaptive moment estimation.

**Learning rate schedules and warm restarts.** Cosine annealing with warm restarts introduced periodic learning rate cycles to prevent premature convergence, allowing the optimizer to escape sharp basins by periodically spiking the learning rate. Tango's explore phase draws inspiration from this idea but combines cyclic learning rates with tangentplane perturbations, so that each highLR excursion is accompanied by structured lateral exploration rather than gradientonly movement. The exploit phase then applies standard cosine decay, preserving the wellunderstood convergence behavior of modern LLM training schedules.

**Sharpnessaware minimization.** SAM explicitly seeks parameters whose entire neighborhood has uniformly low loss, by computing an additional gradient at a worstcase perturbed point before the descent step. This doubles perstep compute but provides strong theoretical guarantees on flatminimum seeking. ASAM extended this with adaptive perturbation scaling. Tango pursues a related goal — biasing training toward flat minima — but achieves it implicitly: the sharpness gate suppresses tangent steps near sharp ridges, and the curvatureinverse scaling in Equation (9) makes tangent steps naturally larger in flat regions. This costs one additional forward pass per S steps rather than a full extra backward pass, giving Tango a substantially lower overhead than SAMfamily methods.

**Loss landscape geometry.** Visualizing the loss landscape of neural networks established that sharp minima correlate with poor generalization while flat minima tend to generalize well — providing the theoretical foundation for sharpnessaware training. The observation that loss surfaces in highdimensional spaces contain exponentially many saddle points rather than local minima motivates escape mechanisms beyond gradient descent. Tango's tangent steps directly address this: by exploring the (D−1)dimensional subspace orthogonal to the gradient, they provide structured escape directions from saddle points and flat ridges that gradientonly methods cannot access.

**Exploration in parameter space.** Stochastic gradient descent with momentum provides implicit exploration through gradient noise from minibatch sampling, but this noise is unstructured and isotropic. Attempts to add structured noise — such as entropySGD, which explicitly maximizes local loss surface entropy — show that deliberate exploration of the loss landscape improves generalization. Tango's tangent momentum mechanism is related in spirit but geometrically grounded: the projection onto the tangent plane ensures exploration is strictly orthogonal to the descent direction, avoiding interference with gradientdriven convergence while covering directions that gradient noise rarely reaches in highdimensional spaces.

## 6. Conclusion

We presented Tango, an optimizer that augments AdamW with a twophase learning rate schedule, a decaying exploration factor φ(t), and sharpnessgated tangent momentum steps operating in the (D−1)dimensional subspace orthogonal to the gradient. A geometric argument shows that Tango’s exploration advantage over gradientonly methods grows linearly with the parameter space dimension, directly reflected in the Rosenbrock results where the improvement factor grows from 28,500× at D=10 to 102,900× at D=50. A posthoc ablation isolates the contribution of the exploration factor decay, finding it essential: without φ(t) decay, tangent exploration prevents convergence and TangoNoEF is outperformed by AdamBest at D≥20 by up to seven orders of magnitude. With all mechanisms active, Tango wins all 15 Rosenbrock seeds. Across MNIST classification, Rosenbrock optimisation, characterlevel language modelling on Shakespeare, and ResNet18 image classification on CIFAR10, Tango outperforms Adam and its cosinedecay variant without a meaningful increase in perstep computational cost. On WikiText2 — the sole benchmark where the cyclic explore phase is replaced by a flat cosine schedule (a domainappropriate adaptation) and no LR retuning was performed — Tango trails the best AdamW configuration by 0.003 validation loss, a gap attributable to LR mismatch rather than a failure of the tangent mechanism. Code and benchmark scripts are publicly released.

## References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[Hochreiter, S., & Schmidhuber, J. (1997). Flat minima. Neural Computation, 9(1), 1–42.]

[Li, H., Xu, Z., Taylor, G., Studer, C., & Goldstein, T. (2018). Visualizing the loss landscape of neural nets. NeurIPS 2018.]

[Karpathy, A. nanoGPT. https://github.com/karpathy/nanoGPT]

# Appendices

## Appendix A – Benchmark Hyperparameter Configurations

For MNIST classification:

| Parameter | Value | Status |
|---|---|---|
| Exploration Learning rate (start) | $3\times10^{-4}$ | Tuned |
| Exploration Learning rate (end) | $3\times10^{-5}$ | Tuned |
| Exploitation Learning rate (start) | $3\times10^{-4}$ | Tuned |
| Exploitation Learning rate (end) | $3\times10^{-5}$ | Tuned |
| Cyclic exploration | 0.80 | Tuned |
| FD step size | $1\times10^{-4}$ | Fixed |
| Sharpness probe frequency | Every 200 steps | Fixed |
| Momentum coefficient | 0.80 | Fixed |
| Tangent step size | $4\times10^{-4}$ | Fixed |
| Tangent step frequency | Every 100 steps | Fixed |
| Noise injection | $\sigma = 0.01$ | Enabled |

For Shakespeare GPT:

| Parameter | Value | Status |
|---|---|---|
| Exploration Learning rate (start) | $3\times10^{-4}$ | Tuned |
| Exploration Learning rate (end) | $3\times10^{-5}$ | Tuned |
| Exploitation Learning rate (start) | $3\times10^{-4}$ | Tuned |
| Exploitation Learning rate (end) | $3\times10^{-5}$ | Tuned |
| Cyclic exploration | — | Disabled |
| FD step size | $1\times10^{-4}$ | Fixed |
| Sharpness probe frequency | Every 200 steps | Fixed |
| Momentum coefficient | 0.80 | Fixed |

| | | |
|---|---|---|
| Tangent step size | $4\times10^{-4}$ | Fixed |
| Tangent step frequency | Every 100 steps | Fixed |
| Noise injection | — | Disabled |

For Rosenbrock Function:

| Parameter | Value | Status |
|---|---|---|
| Exploration Learning rate (start) | $3\times10^{-4}$ | Tuned |
| Exploration Learning rate (end) | $3\times10^{-5}$ | Tuned |
| Exploitation Learning rate (start) | $3\times10^{-4}$ | Tuned |
| Exploitation Learning rate (end) | $3\times10^{-5}$ | Tuned |
| Cyclic exploration | 0.80 | Tuned |
| FD step size | $1\times10^{-4}$ | Fixed |
| Sharpness probe frequency | Every 200 steps | Fixed |
| Momentum coefficient | 0.80 | Fixed |
| Tangent step size | $4\times10^{-4}$ | Fixed |
| Tangent step frequency | Every 100 steps | Fixed |
| Noise injection | — | Disabled |

For WikiText2 GPT :

| Parameter | Value | Status |
|---|---|---|
| LR (explore start) | $3\times10^{-4}$ | Tuned |
| LR (explore end) | $1\times10^{-5}$ | Tuned |
| LR (exploit start) | $3\times10^{-4}$ | Tuned |
| LR (exploit end) | $1\times10^{-5}$ | Tuned |
| Cyclic exploration | 0.30 | Fixed |
| FD step size | $1\times10^{-4}$ | Fixed |
| Sharpness probe frequency | Every 200 steps | Fixed |
| Momentum coefficient | 0.80 | Fixed |
| Tangent step size | $4\times10^{-4}$ | Fixed |
| Tangent step frequency | Every 100 steps | Fixed |
| Noise injection | — | Disabled |

For CIFAR10 / ResNet18 (best preliminary LR selected at 1500 steps, then run for full training):

| Parameter | Value | Status |
|---|---|---|
| LR grid searched (5 LRs) | $10^{-2}$, $3\times10^{-3}$, $10^{-3}$, $3\times10^{-4}$, $10^{-4}$ | Tuned |
| Preliminary selection steps | 1,500 | Fixed |
| FD step size | $1\times10^{-4}$ | Fixed |
| Momentum coefficient | 0.80 | Fixed |
| Tangent step size | $4\times10^{-4}$ | Fixed |
| Noise injection | — | Disabled |

## Appendix B – PerSeed Results

For MNIST benchmark:

| MNIST (Over 5 seeds, 3 optimizers) | | | | | |
|---|---|---|---|---|---|
| Seed | Tango | LR | Adamfixed | Adamcosine | Winner |
| 0 | 0.0251 | 1.00E02 | 0.0486 | 0.0439 | Tango |
| | | 3.00E03 | 0.0323 | 0.0295 | |
| | | 1.00E03 | 0.0275 | 0.0267 | |
| | | 3.00E04 | 0.0324 | 0.0334 | |
| | | 1.00E04 | 0.0433 | 0.0527 | |
| 1 | 0.023 | 1.00E02 | 0.0561 | 0.0473 | Tango |
| | | 3.00E03 | 0.0325 | 0.028 | |
| | | 1.00E03 | 0.0287 | 0.0249 | |
| | | 3.00E04 | 0.0302 | 0.0339 | |
| | | 1.00E04 | 0.041 | 0.0519 | |
| 2 | 0.0206 | 1.00E02 | 0.0432 | 0.0335 | Tango |
| | | 3.00E03 | 0.0336 | 0.0264 | |
| | | 1.00E03 | 0.028 | 0.0274 | |
| | | 3.00E04 | 0.0294 | 0.0312 | |
| | | 1.00E04 | 0.0434 | 0.0519 | |
| 3 | 0.0253 | 1.00E02 | 0.0493 | 0.0518 | Adam Cosine |
| | | 3.00E03 | 0.0293 | 0.0276 | |
| | | 1.00E03 | 0.0281 | 0.0245 | |
| | | 3.00E04 | 0.0293 | 0.0343 | |
| | | 1.00E04 | 0.0485 | 0.0586 | |
| 4 | 0.0228 | 1.00E02 | 0.0603 | 0.0431 | Tango |
| | | 3.00E03 | 0.0304 | 0.0288 | |
| | | 1.00E03 | 0.0329 | 0.0293 | |
| | | 3.00E04 | 0.0316 | 0.034 | |
| | | 1.00E04 | 0.0442 | 0.059 | |

For ShakespeareGPT benchmark:

| Seed | Tango | | Adam | | | Winner |
|---|---|---|---|---|---|---|
| | val loss | ppl | LR | val loss | ppl | |
| 0 | 1.4655 | 4.33 | 1.00E02 | 2.6116 | 13.62 | Tango |
| | | | 3.00E03 | 2.681 | 14.6 | |
| | | | 1.00E03 | 1.6245 | 5.08 | |
| | | | 3.00E04 | 1.5193 | 4.57 | |
| | | | 1.00E04 | 1.501 | 4.49 | |
| 1 | 1.4547 | 4.28 | 3.00E04 | 1.54 | 4.66 | Tango |
| | | | 1.00E04 | 1.505 | 4.5 | |
| 2 | 1.4871 | 4.42 | 3.00E04 | 1.5534 | 4.73 | Tango |
| | | | 1.00E04 | 1.5139 | 4.54 | |
| 3 | 1.4625 | 4.32 | 3.00E04 | 1.5375 | 4.65 | Tango |
| | | | 1.00E04 | 1.4916 | 4.44 | |

For Rosenbrock Function:

| Dimension | Seed | Tango | Adam (best) | Winner | Wall time |
|---|---|---|---|---|---|
| 10 | 0 | 0 | 8.72E11 | Tango | 95.6s |
| | 1 | 1.65E12 | 1.69E10 | Tango | 86.0s |
| | 2 | 5.02E12 | 5.55E07 | Tango | 89.0s |
| | 3 | 0 | 9.76E11 | Tango | 120.5s |
| | 4 | 1.28E11 | 1.51E10 | Tango | 42.7s |
| | mean | 3.90E12 | 1.11E07 | 5/5 Tango | — |
| | std | 4.82E12 | 2.22E07 | ratio $2.85\times10^4$ · +4.45 OoM | |
| 20 | 0 | 0 | 7.75E09 | Tango | 30.1s |
| | 1 | 3.73E13 | 2.76E12 | Tango | 31.0s |
| | 2 | 3.59E13 | 4.97E10 | Tango | 31.3s |
| | 3 | 4.63E11 | 7.00E09 | Tango | 31.4s |
| | 4 | 5.05E11 | 9.35E08 | Tango | 37.6s |
| | mean | 1.95E11 | 2.18E08 | 5/5 Tango | — |
| | std | 2.36E11 | 3.60E08 | ratio $1.11\times10^3$ · +3.05 OoM | |
| 50 | 0 | 1.15E10 | 2.30E05 | Tango | 31.5s |
| | 1 | 1.21E10 | 1.82E05 | Tango | 33.1s |
| | 2 | 1.07E10 | 2.37E07 | Tango | 47.4s |
| | 3 | 1.82E11 | 0 | Adam | 33.9s |
| | 4 | 4.16E11 | 4.26E10 | Tango | 80.5s |
| | mean | 8.05E11 | 8.28E06 | 4/5 Tango | — |
| | std | 4.22E11 | 1.02E05 | ratio $1.03\times10^5$ · +5.01 OoM | |

## Appendix C – FLOPS Derivation

We derive the perstep computational cost of Tango relative to a standard AdamW baseline to justify the overhead figures reported in Section 4.3.

**Baseline cost (AdamW)**
A single AdamW gradient step requires one forward pass and one backward pass. Following standard convention, the backward pass costs approximately twice the forward pass. Thus:
Cost per AdamW step = $3 \times F$
where F denotes the cost of a single forward pass through the model.

**Tango additional costs**
Tango adds two mechanisms on top of the base AdamW step:
*Sharpness probe (every S = 200 step):* One additional forward pass to evaluate $\mathcal{L}(\theta + \varepsilon\dot{g})$. No backward pass is required — the gradient $\dot{g}$ is reused from the main step. Amortised over S steps this adds:
$1/S \times F = 0.005 \times F$ per step
*Tangent step (every N = 100 step, when gates pass):* The tangent perturbation is computed from the cached gradient — no additional forward or backward pass is needed. The parameter update itself is a vector addition of negligible cost. When the gates fail (sharpness or loss gate not satisfied), cost is exactly zero.
Amortised tangent cost ≈ 0 additional forward passes per step

**Total Tango cost per step:** $3F + (1/200)\,F = 3.005F$

**Overhead relative to baseline:** $(3.005F - 3F) / 3F = 0.005/3 \approx 0.167\%$

**Empirical overhead (Damped Oscillatory Regression):**
The theoretical figure of ~0.17% represents a lower bound. Empirically measured overhead is slightly higher due to:

- Python loop overhead for the gate checks
- Memory operations for parameter save/restore during the FD probe
- Gradient extraction for the tangent projection

Measured empirically across benchmarks:

| Benchmark | Baseline (TFLOPs) | Tango (TFLOPs) | Computational Overhead(%) |
|---|---|---|---|
| **ShakespeareGPT** | 6,840.7 | 6,859.4 | 0.27 |
| **Rosenbrock Function** | 1.424e5 | 1.420e5 | 0.31 |
| **Wikitext2** | 2,914 | 2,923 | 0.32 |
| **MNIST** | 4.7445 | 4.7371 | 0.20 |
| **ResNet18** | 4168 | 4182 | 0.33 |

The decreasing overhead at larger model scale confirms the theoretical prediction: as model size grows, the fixed cost of the FD probe (one forward pass every 200 steps) becomes negligible relative to the dominant cost of the gradient step itself.

## Appendix D — Convergence Analysis of Tango

We prove that the sequence of iterates produced by Tango visits a firstorder stationary point in the limit, and that the bestcheckpoint parameter $\theta^*$ satisfies $\nabla\mathcal{L}(\theta^*) \to 0$ as $T \to \infty$.

### D.1 Assumptions

**Assumption 1 (LSmoothness).** $\mathcal{L} : \mathbb{R}^D \to \mathbb{R}$ is continuously differentiable and its gradient is Lipschitz continuous with constant $L > 0$:
$\|\nabla\mathcal{L}(\theta_1) - \nabla\mathcal{L}(\theta_2)\| \leq L \|\theta_1 - \theta_2\|, \forall \theta_1, \theta_2 \in \mathbb{R}^D$.

**Assumption 2 (Lower Boundedness).** $\mathcal{L}(\theta) \geq \mathcal{L}^* > -\infty$ for all $\theta$.

**Assumption 3 (Bounded Gradients).** There exists $G < \infty$ such that $\|\nabla\mathcal{L}(\theta_t)\| \leq G$ for all t.
**Assumption 4 (Positive Curvature at Attempted Steps).** A tangent step is only attempted at step t if the curvature estimate satisfies $\kappa_t > 0$. (When $\kappa_t \leq 0$ the region is flat or locally concave along the gradient direction; Tango suppresses the attempt via the sharpness gate, so no tangent step is proposed.)

**Assumption 5 (Decaying Learning Rate).** The base AdamW learning rate satisfies $\eta_t \leq 1/L$ for all t. This holds under the cosine schedule once a sufficiently small LR_EXPLOIT_END is chosen, and can be enforced as a warmupindependent sufficient condition.

**Assumption 6 (ρsmooth Hessian):** The Hessian $\nabla^2\mathcal{L}$ is Lipschitz continuous with constant $\rho > 0$ $\|\nabla^2\mathcal{L}(\theta_1) - \nabla^2\mathcal{L}(\theta_2)\| \leq \rho\|\theta_1 - \theta_2\|$. This is standard in thirdorder Taylor analyses and holds for all smooth neural network objectives with bounded weights.

**Remark.** Assumption 3 is standard for stochastic objectives and is satisfied empirically by gradient clipping (clip norm = 1.0) used in all Tango GPT benchmarks. In the deterministic Rosenbrock setting it holds trivially on any compact sublevel set.

**D.2 Structure of a Single Tango Step**
Each iteration t decomposes into two substeps:

1. **Base step.** Compute gradient $g_t = \nabla\mathcal{L}(\theta_t)$, apply AdamW:

$\theta_{t+½} = \theta_t - \eta_t \hat{g}_t$
where $\hat{g}_t$ is the AdamW preconditioned direction ($\hat{g}_t = g_t/\|g_t\|$ in the simplified SGD case; the adaptive case is handled in Remark D.4).

2. **Tangent step (conditional).** If t is a tangentstep index (t mod N = 0, t > t_warmup) and $\kappa_t > 0$ and $\mathcal{L}(\theta_t) > \mathcal{L}$_best:
    1. (a) Draw direction $d_t$ from the tangent plane: $g_t^\top d_t = 0$, $\|d_t\| = 1$.
    2. (b) Set $\delta_t = \varepsilon_t d_t$ where $\varepsilon_t = \varepsilon_0 \sqrt{(\sigma/|\kappa_t|)} \cdot \varphi(t)$, clipped to $[\varepsilon\_min, \Delta_t]$.
    3. (c) Compute $\rho_t = (\mathcal{L}(\theta_{t+½}) - \mathcal{L}(\theta_{t+½} + \delta_t)) / m_t(\delta_t)$, where $m_t(\delta_t) = ½ |\kappa_t| \|\delta_t\|^2$ (the $g_t^\top\delta_t = 0$ orthogonality causes the linear term to vanish exactly).
    4. (d) If $\rho_t \geq \eta_0$ (accepted): $\theta_{t+1} = \theta_{t+½} + \delta_t$; expand $\Delta_t$ if $\rho_t \geq \eta_1$.
    5. (e) If $\rho_t < \eta_0$ (rejected): $\theta_{t+1} = \theta_{t+½}$; contract $\Delta_t$.
    6. Otherwise (gates fail or nontangent step): $\theta_{t+1} = \theta_{t+½}$.

**D.3 Proof of Convergence**
**Step A — Sufficient Decrease from the Base Step**
By the Lsmoothness descent lemma applied to the base step:
$\mathcal{L}(\theta_{t+½}) \leq \mathcal{L}(\theta_t) + \eta_t\, g_t^\top(-\hat{g}_t) + (L\eta_t^2/2) \|\hat{g}_t\|^2$
For the simplified SGD case ($\hat{g}_t = \eta_t\, g_t$):
$\mathcal{L}(\theta_{t+½}) \leq \mathcal{L}(\theta_t) - \eta_t \|g_t\|^2 + (L\eta_t^2/2) \|g_t\|^2$
$= \mathcal{L}(\theta_t) - \eta_t(1 - L\eta_t/2) \|g_t\|^2$
Under Assumption 5 ($\eta_t \leq 1/L$), the factor $(1 - L\eta_t/2) \geq ½$, giving:
$\mathcal{L}(\theta_{t+½}) - \mathcal{L}(\theta_t) \leq -(\eta_t/2) \|\nabla\mathcal{L}(\theta_t)\|^2 \equiv -c_1 \|\nabla\mathcal{L}(\theta_t)\|^2$ (A1)
where $c_1 = \eta_t/2 > 0$. (See Remark D.4 for the AdamW extension.)
**Step B — NonAscent from the Tangent Step**
**Case B1: tangent step rejected ($\rho_t < \eta_0$).** $\theta_{t+1} = \theta_{t+½}$. No change to the loss; (A1) holds without modification.
**Case B2: tangent step accepted ($\rho_t \geq \eta_0$).** By definition of $\rho_t$:
$\rho_t = (\mathcal{L}(\theta_{t+½}) - \mathcal{L}(\theta_{t+½} + \delta_t)) / m_t(\delta_t) \geq \eta_0$
Since $m_t(\delta_t) = ½ |\kappa_t| \|\delta_t\|^2 > 0$ (Assumption 4), rearranging gives the **tangent decrease bound**:

$\mathcal{L}(\theta_{t+½}) - \mathcal{L}(\theta_{t+½} + \delta_t) \geq \eta_0 \cdot ½ |\kappa_t| \|\delta_t\|^2$ (B1)
Hence $\mathcal{L}(\theta_{t+1}) = \mathcal{L}(\theta_{t+½} + \delta_t) \leq \mathcal{L}(\theta_{t+½})$, so the tangent step never increases the loss above the basestep iterate.
**Combined bound for any t:**
$\mathcal{L}(\theta_{t+1}) - \mathcal{L}(\theta_t) \leq -c_1 \|\nabla\mathcal{L}(\theta_t)\|^2 - \mathbb{1}[t \in T_acc] \cdot (\eta_0 |\kappa_t|/2) \|\delta_t\|^2$ (C1)
where $T_acc \subseteq \mathbb{N}$ is the (random) set of accepted tangentstep indices. Both terms on the right are nonpositive, confirming monotone descent along the combined trajectory.
**Step C — Telescoping and Convergence**
Summing (C1) from t = 0 to T:
$\mathcal{L}(\theta_{T+1}) - \mathcal{L}(\theta_0) \leq -c_1 \Sigma_{t=0}^T \|\nabla\mathcal{L}(\theta_t)\|^2 - \Sigma_{t \in T_acc} (\eta_0|\kappa_t|/2) \|\delta_t\|^2$
Since both subtracted terms are nonnegative and $\mathcal{L}(\theta_{T+1}) \geq \mathcal{L}^*$ (Assumption 2):
$c_1 \Sigma_{t=0}^T \|\nabla\mathcal{L}(\theta_t)\|^2 \leq \mathcal{L}(\theta_0) - \mathcal{L}^* < \infty$
Taking $T \to \infty$, the series converges absolutely. Since $c_1 > 0$, the individual terms must vanish:
**$\lim_{t\to\infty} \|\nabla\mathcal{L}(\theta_t)\| = 0$ □**
The sequence $\{\theta_t\}$ visits a neighbourhood of a firstorder stationary point infinitely often.

**D.4 The BestCheckpoint and Cyclic LR: lim inf Guarantee**
Tango maintains $\theta^* = \text{argmin}_t\ \mathcal{L}(\theta_t)$. Because $\{\mathcal{L}(\theta_t)\}$ is nonincreasing at accepted steps (established above by the combined descent bound (C1)), the best checkpoint $\theta^*$ satisfies $\mathcal{L}(\theta^*) \leq \mathcal{L}(\theta_t)$ for all subsequent t. From the telescoping bound:
$\|\nabla\mathcal{L}(\theta_t)\| \to 0 \Longrightarrow \liminf_{t\to\infty} \|\nabla\mathcal{L}(\theta_t)\| = 0$
By continuity of $\nabla\mathcal{L}$ and the fact that $\theta^*$ is attained at a finite index $t^*$ where $\|\nabla\mathcal{L}(\theta_{t^*})\|$ is within ε of zero, the checkpoint satisfies:
**$\|\nabla\mathcal{L}(\theta^*)\| \leq \varepsilon$ for any $\varepsilon > 0$ chosen sufficiently small.**
**Cyclic LR (TangoFull).** When the cyclic explore phase is active ($t <$ EXPLORE_FRAC · T), the learning rate oscillates and $\{\mathcal{L}(\theta_t)\}$ is not monotone. However, bestcheckpoint tracking ensures $\mathcal{L}(\theta^*)$ is nonincreasing across cycles. Formal convergence for this case follows Loshchilov & Hutter (2017, SGDR), who establish that SGD with cyclic restarts satisfies $\liminf \|\nabla\mathcal{L}(\theta_t)\| = 0$ under Lsmoothness and lower boundedness. Since Tango's tangent steps are accepted only when they decrease the loss (B2), they do not invalidate this guarantee; they can only accelerate escape from local basins during the cyclic phase.

**D.5 The Curvature Proxy: Validity of the Scalar Model**

**Lemma D.5.1 (Explicit remainder bound).** Under Assumption 6, for any δ with $g^T\delta = 0$:
$|\mathcal{L}(\theta + \delta) - \mathcal{L}(\theta) - ½ \delta^T H\delta| \leq (\rho/6)\|\delta\|^3$
**Proof.** Direct application of Taylor's theorem with integral remainder under Lipschitz Hessian. □
**Condition for proxy validity.** The scalar model $m_t(\delta_t) = ½|\kappa_t|\|\delta_t\|^2$ uses $\kappa_t = \hat{g}_t^T H\hat{g}_t$ as a proxy for $d_t^T H d_t$. The relative error of the quadratic approximation satisfies:
$|\text{remainder}| / |\text{quadratic term}| \leq (\rho/6)\|\delta_t\|^3 / (|d_t^T H d_t|/2 \cdot \|\delta_t\|^2) = \rho\|\delta_t\| / (3|d_t^T H d_t|)$
This ratio $\to 0$ as $\|\delta_t\| \to 0$, and is strictly less than 1 when:
$\|\delta_t\| < 3|d_t^T H d_t| / \rho$ ... (*)
**Explicit bound on $\|\delta_t\|$.** From the step size formula:
$\|\delta_t\| \leq \varepsilon_t = \varepsilon_0 \sqrt{(\sigma/|\kappa_t|)} \cdot \varphi(t) \leq \varepsilon_0 \sqrt{(\sigma/\kappa_{min})} \cdot \varphi_0$
where $\kappa_{min} = \inf_t |\kappa_t| > 0$ (Assumption 4). Condition (*) is therefore satisfied when:
$\varepsilon_0 \sqrt{(\sigma/\kappa_{min})} \cdot \varphi_0 < 3\lambda_{min}(H) / \rho$
This gives an **explicit formula for a valid $\varphi_0$**:
$\varphi_0 < (3\lambda_{min}(H) / \rho) \cdot (1 / \varepsilon_0 \sqrt{(\sigma/\kappa_{min})})$

**Issue.** The trustregion model $m_t(\delta_t) = ½ |\kappa_t| \|\delta_t\|^2$ uses $\kappa_t \approx \hat{g}_t^T H \hat{g}_t$ (the Rayleigh quotient along the gradient direction), while the tangent step $\delta_t$ is orthogonal to $\hat{g}_t$. The curvature along $\delta_t$ is the eigenvalue of H in the direction $d_t$, which may differ from $\kappa_t$.
**Resolution via the Exploration Factor Decay φ(t).** Since $\varepsilon_t = \varepsilon_0 \sqrt{(\sigma/|\kappa_t|)} \cdot \varphi(t)$ and $\varphi(t) \to \varphi_0$ as $t \to T$, the trust radius collapses as:
$\|\delta_t\| \leq \varepsilon_t \leq \varepsilon_0 \sqrt{(\sigma/|\kappa_t|)} \cdot \varphi_0 \to 0$ as $\varphi_0 \to 0$.

By Lsmoothness, for any direction d with $\|d\| = \varepsilon \to 0$:
$\mathcal{L}(\theta + \varepsilon d) = \mathcal{L}(\theta) + \varepsilon\, g^\top d + (\varepsilon^2/2)\, d^\top H d + O(\varepsilon^3)$
Since $g^\top \delta_t = 0$ exactly, the $O(\varepsilon)$ term vanishes and:
$\mathcal{L}(\theta_{t+½} + \delta_t) - \mathcal{L}(\theta_{t+½}) = (\|\delta_t\|^2/2)\, d_t^\top H d_t + O(\|\delta_t\|^3)$
For sufficiently small $\varphi_0$ ($\varphi_0 = 0.03$ in all experiments), $\|\delta_t\|$ is small enough that $O(\|\delta_t\|^3) \ll O(\|\delta_t\|^2)$, and $d_t^\top H d_t$ is bounded above by $\lambda\_max(H) \le L$ (by Lsmoothness). The scalar proxy $\kappa_t$ therefore serves as a conservative upper bound on the actual curvature along $\delta_t$ in this regime, and the model $m_t$ remains a valid (over)prediction of the reduction. The $\rho_t$ acceptance gate is therefore conservative: it may reject some beneficial steps (since the actual curvature could be smaller than $\kappa_t$), but never accepts a step that truly increases the loss.
**Corollary.** Under the conditions above, the acceptance gate $\rho_t \ge \eta_0$ is both sufficient and conservative: no accepted tangent step increases $\mathcal{L}(\theta)$, and the descent bound (B1) holds with $|\kappa_t|$ replaced by any upper bound on the true directional curvature $d_t^\top H d_t$.

**D.6 Remark: Extension to AdamW**
The base step analysis in Step A uses SGD for clarity. For AdamW, the preconditioned direction $\hat{g}_t = v_t$^{½} $\odot$ $g_t$ (elementwise) satisfies
$g_t^\top \hat{g}_t = \Sigma_i\, g_{t,i}^2 / \sqrt{(v_{t,i})} \ge (1/\sqrt{v\_max})\, \|g_t\|^2$
where $v\_max = max\_i\ v_{t,i}$ is bounded above by $G^2$ (Assumption 3) and below by $\varepsilon\_adam > 0$ (the AdamW numerical stabiliser). This gives the modified descent constant $c_1 = \eta_t/(2\sqrt{v\_max}) > 0$, and the telescoping argument in Step C proceeds identically. A detailed proof using the AdamWspecific geometry is given in Défossez et al. (2022), whose convergence result for Adam under Lsmoothness and bounded gradients applies directly to Tango's base step.
**D.7 Summary**

| Claim | Established |
|---|---|
| Base step guarantees sufficient decrease | Step A, Eq. (A1) |
| Accepted tangent step never increases loss | Step B, Eq. (B1) |
| Rejected tangent step is a noop | Case B1 |
| Gradient norms vanish along trajectory | Step C |
| Best checkpoint θ* is nearstationary | Section D.4 |
| Scalar curvature proxy is valid near convergence | Section D.5 |
| Result extends to AdamW base step | Section D.6 |

The primary open question is tightening the rate: the analysis above establishes asymptotic convergence but not an explicit $O(1/\sqrt{T})$ or $O(1/T)$ rate for the full combined system. Establishing a nonasymptotic rate under the cyclic LR phase remains future work.